

MTH 553: Lecture # 38

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May 1, 2026

Lecture Span

1. Reaction diffusion traveling wave

Traveling wave for a bistable reaction diffusion equation

$$u_t = u_{xx} + f(u) \quad (1)$$

Here, $f(u)$ is a reaction term. This models the diffusion of a substance with chemical reaction generating new material at the rate of $f(u)$.

Suppose our curve f has two humps whose middle intersects a . Note that the top hump is larger than the bottom hump. Here, $0 < a < 1$ and

$$\int_0^1 f(z) dz > 0 \quad (2)$$

That is, there is more area over the axis than under the axis. But now, what does bistable mean? Here, $u \equiv 0$ is a stable solution of the PDE.

If $u \equiv -\epsilon$ then $f(u) > 0$, so then u approaches 0. If $u \equiv \epsilon$, then $f(u) < 0$. So then u also approaches 0. This is to say that the solution $u \equiv 0$ is stable.

Similarly, $u \equiv 1$ is also stable. However, $u = a$ is unstable. Therefore, we call the PDE bistable because there are two stable steady state solutions.

Our goal is find

$$u(x, t) = v(x + ct) \quad (3)$$

We need that

$$\begin{cases} v(-\infty) = 0 \\ v'(\pm\infty) = 0 \end{cases} \quad (4)$$

Now, substitute into the PDE to obtain

$$cv'(s) = v''(s) + f(v(s)) \quad (5)$$

Where $s = x + ct \in \mathbb{R}$. Here, we will use the phase plane method to turn this 2nd order ODE into two first order ODEs. So then, we obtain

$$v' = w \quad (6)$$

$$w' = f(v) + cw \quad (7)$$

Then we want

1. $(v, w) \rightarrow (1, 0)$ as $s \rightarrow \infty$
2. $(v, w) \rightarrow (0, 0)$ as $s \rightarrow -\infty$

Here, to understand the system at $(0,0)$, we aim to linearize the ODE:

$$\begin{pmatrix} v' \\ w' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -f'(0) & c \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix} \quad (8)$$

Then find the eigenvalues of A (the above matrix) to obtain

$$\lambda_{\pm} = \frac{c \pm \sqrt{c^2 - 4f'(0)}}{2} \quad (9)$$

At $(1,0)$, the linearization is

$$\begin{pmatrix} (v-1)' \\ w' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -f'(1) & c \end{pmatrix} \begin{pmatrix} v-1 \\ w \end{pmatrix} \quad (10)$$

Then the eigenvalues of

$$\lambda_{\pm} = \frac{c \pm \sqrt{c^2 - 4f'(1)}}{2} \quad (11)$$

Note, $f'(0) < 0$ and $f'(1) < 0$, which then means that

$$\lambda_- < 0 < \lambda_+ \quad (12)$$

Therefore, you have an incoming eigenvector and an outgoing eigenvector. We claim that there is a unique $c > 0$ such that the outgoing trajectories between $(0,0)$ and $(1,0)$ are connected.

Here, to find such a c , use Lypunov function

$$E(v, w) = \int_0^v f(z)dz + \frac{1}{2}w^2 \quad (13)$$

Which increases along trajectories of ODE systems. Then

$$\frac{d}{ds}E(v(s), w(s)) = (f(v(s)) + v''(s))v'(s) \quad (14)$$

$$= cv'(s)^2 > 0 \quad (15)$$